

(19) **United States**

(12) **Patent Application Publication**
Krishna et al.

(10) **Pub. No.: US 2025/0280323 A1**

(43) **Pub. Date: Sep. 4, 2025**

(54) **METHODS, ARCHITECTURES, APPARATUSES AND SYSTEMS FOR DATA-DRIVEN PREDICTION OF EXTENDED REALITY (XR) DEVICE USER INPUTS BY MULTIPLE DISTRIBUTED DEEP NEURAL NETWORKS (DNNS)**

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(21) Appl. No.: **18/591,733**

(22) Filed: **Feb. 29, 2024**

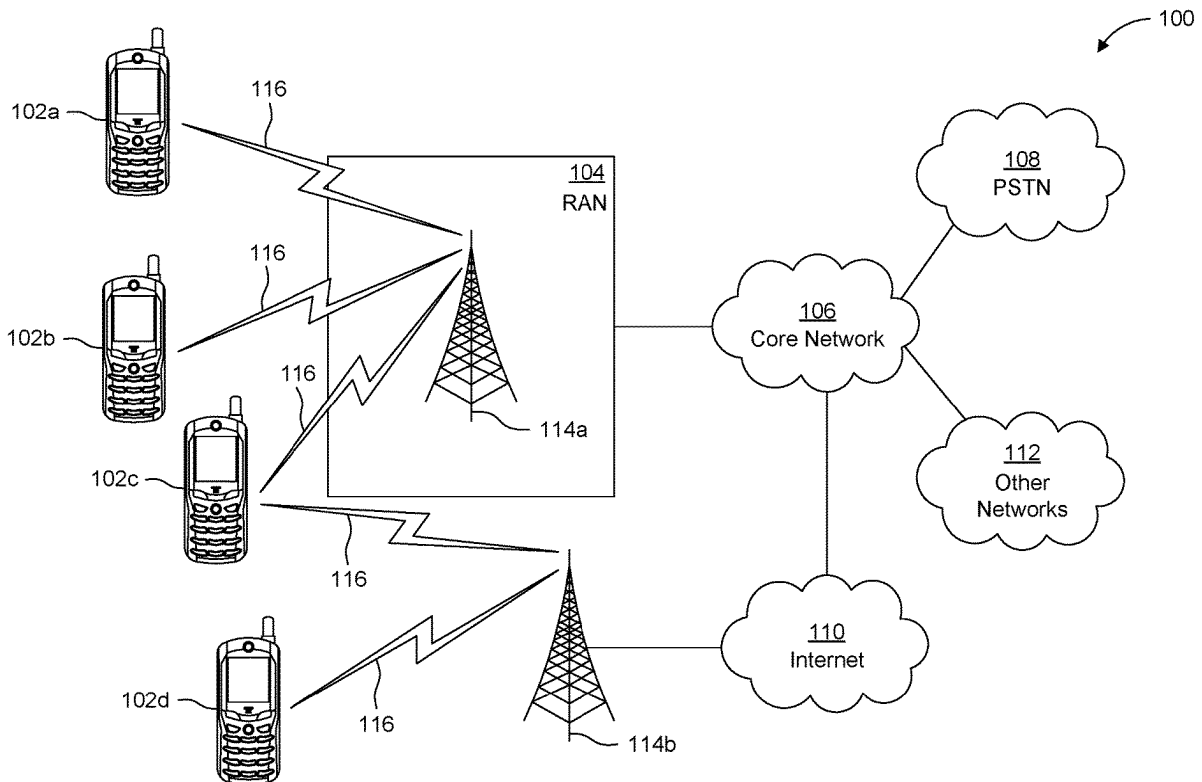
Publication Classification

(51) **Int. Cl.**
H04W 28/04 (2009.01)
H04L 41/147 (2022.01)

(52) **U.S. Cl.**
CPC **H04W 28/04** (2013.01); **H04L 41/147**
(2013.01)

(57) **ABSTRACT**

Procedures, methods, architectures, apparatuses, systems, devices, and computer program products are described for artificial intelligence/machine learning (AIML) models training. For example, a wireless transmit/receive unit (WTRU) is configured to send, to a network entity, first information comprising first data, wherein the network entity comprises a plurality of AI/ML models; receive, from the network entity, second data, wherein the second data are predicted data obtained from an aggregation of output data generated from the plurality of AIML models; determine a prediction error; send, to the network entity, a list of error correction algorithms associated with the plurality of AIML models; receive, from the network entity, second information indicating a rank-ordered list of the error correction algorithms; select an error correction algorithm based on the rank-ordered list of the error correction algorithms; and send, to the network entity, third information indicating the selected error correction algorithm.



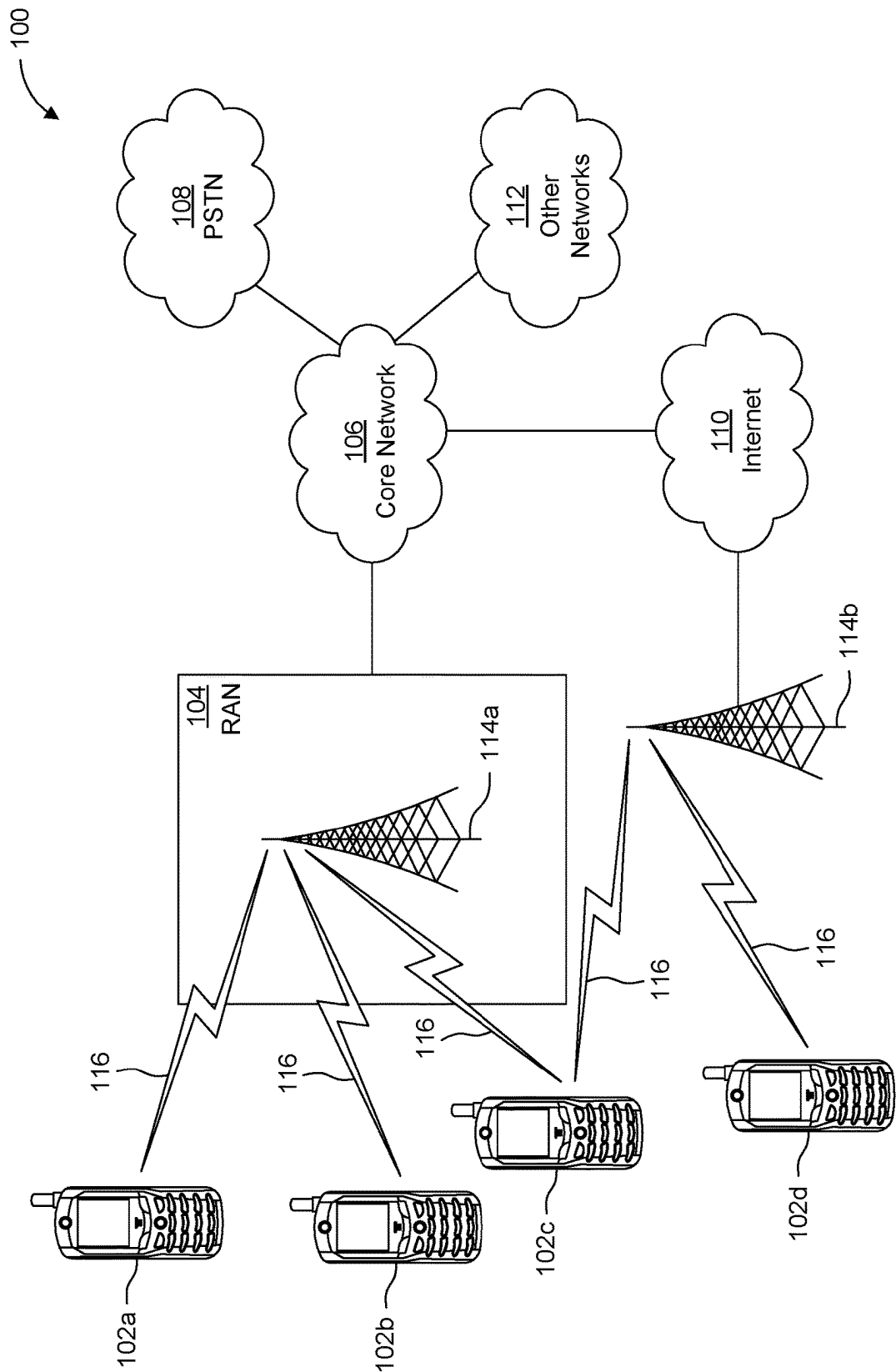


FIG. 1A

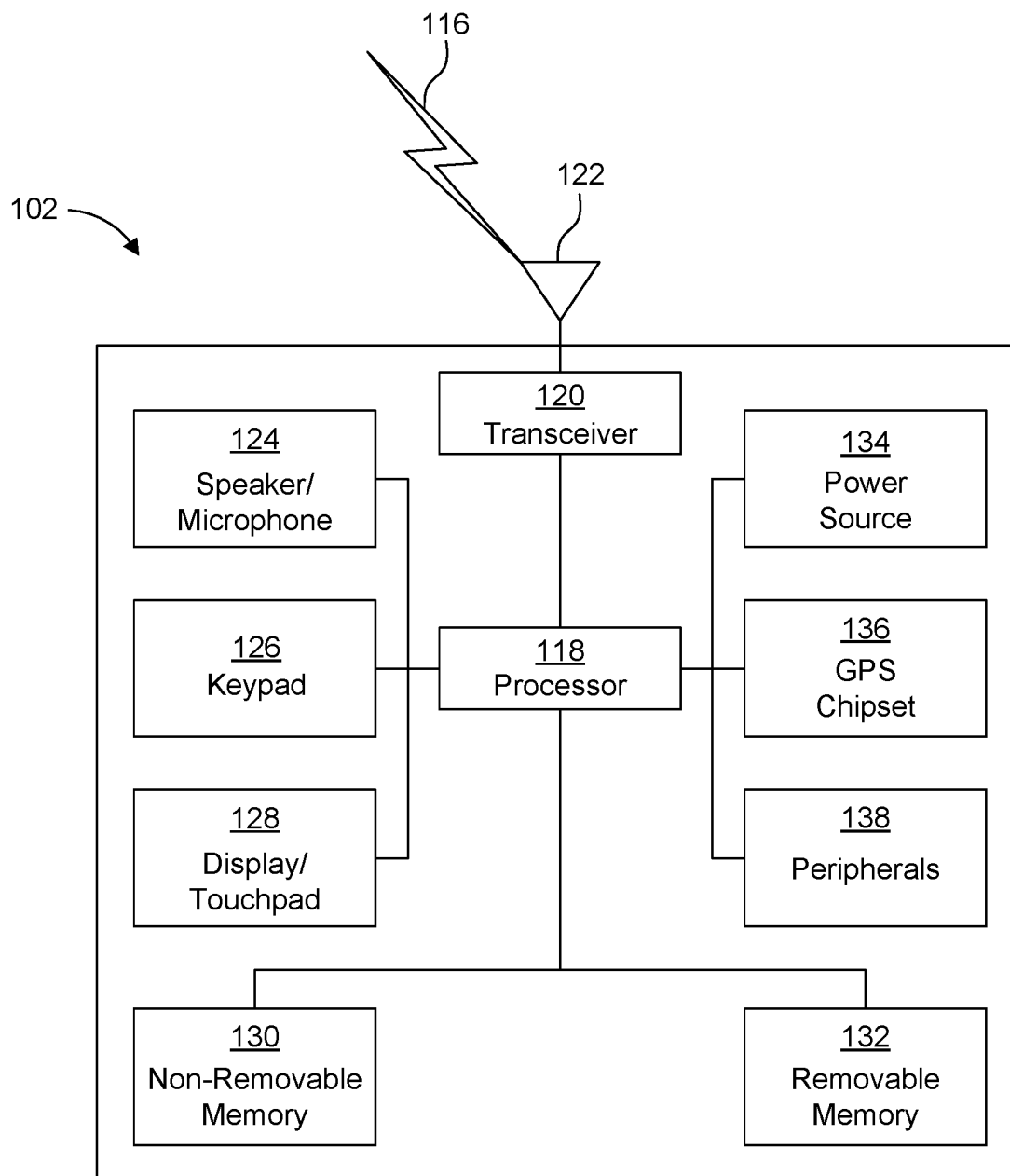


FIG. 1B

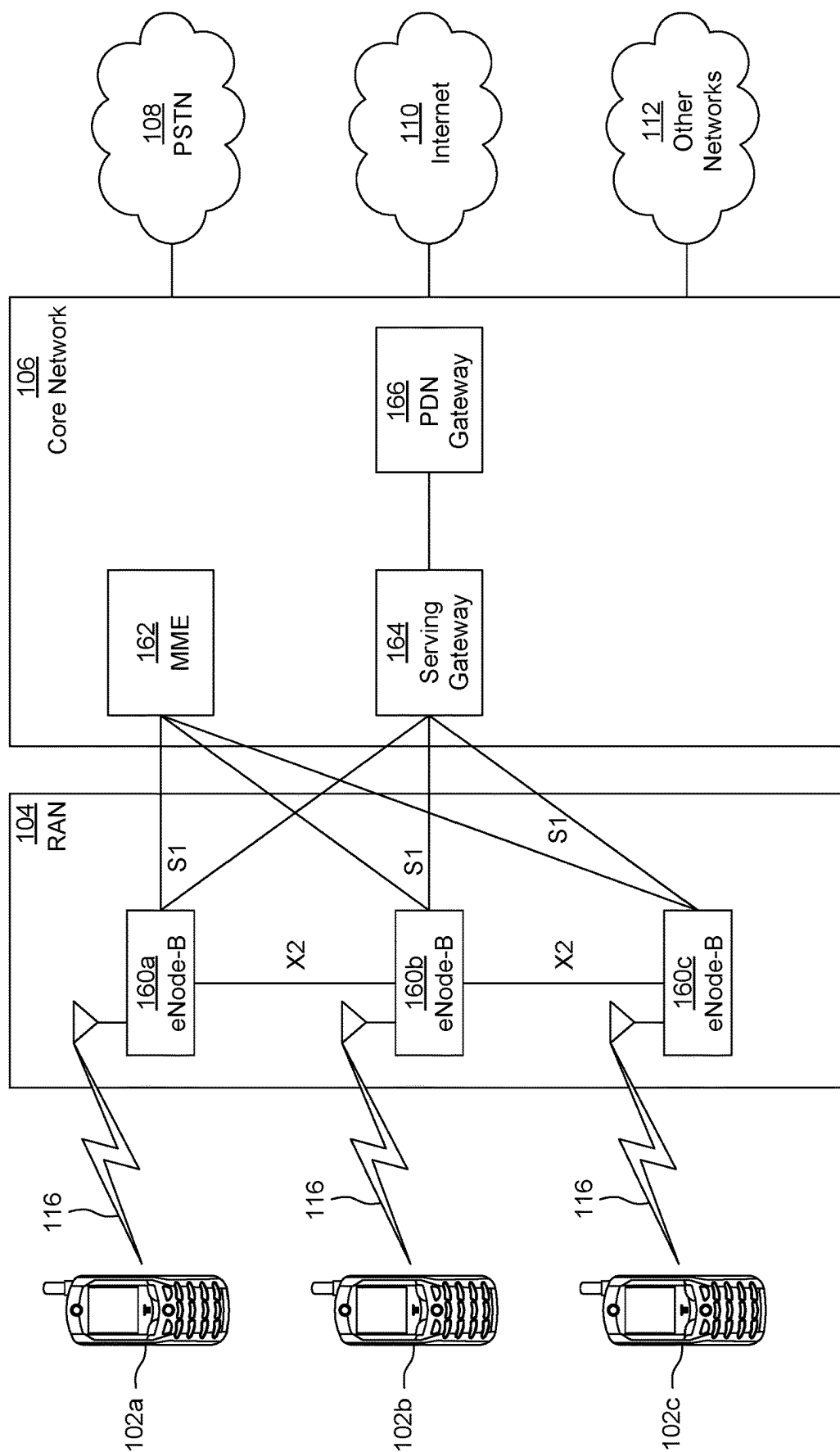


FIG. 1C

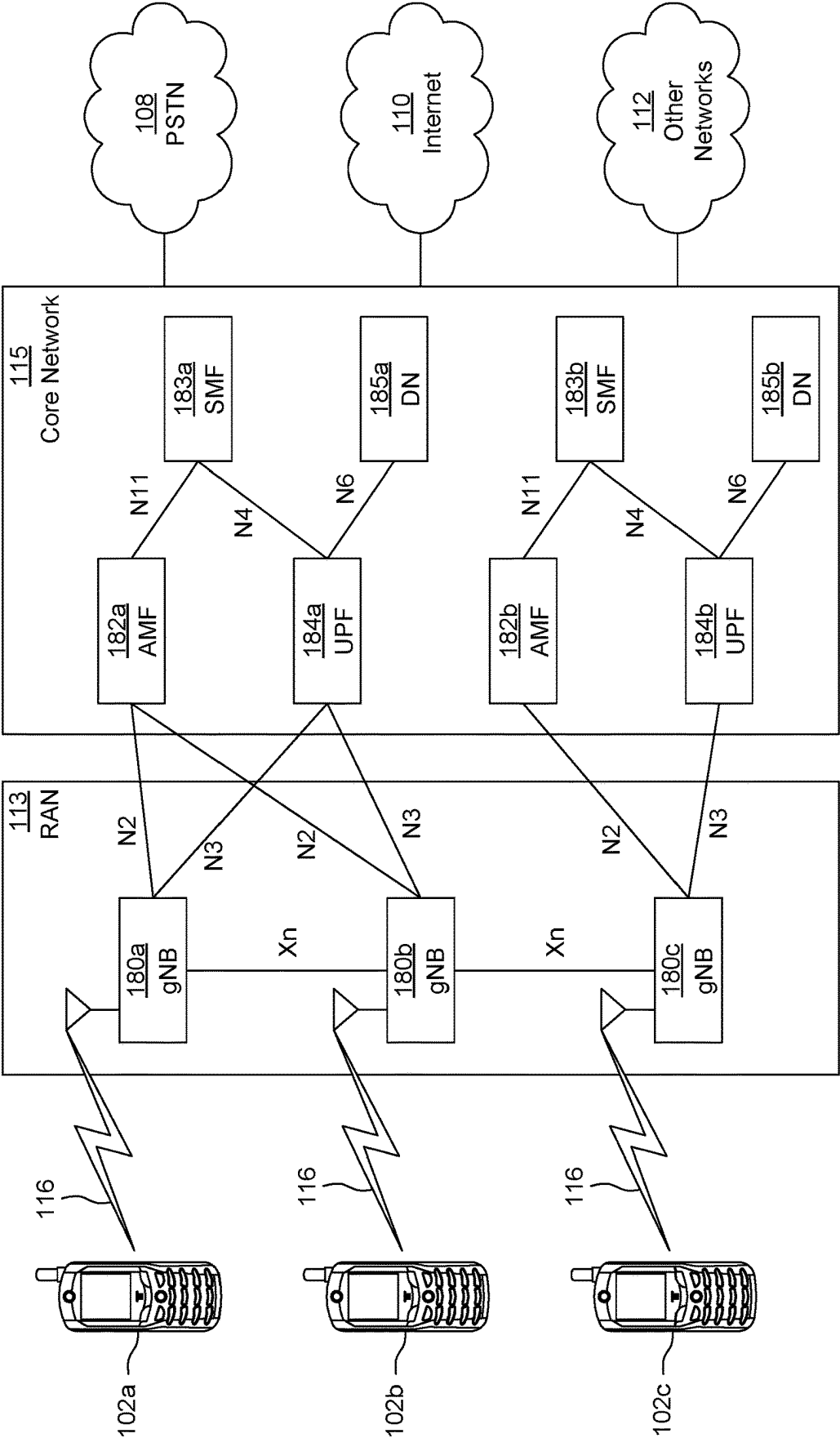


FIG. 1D

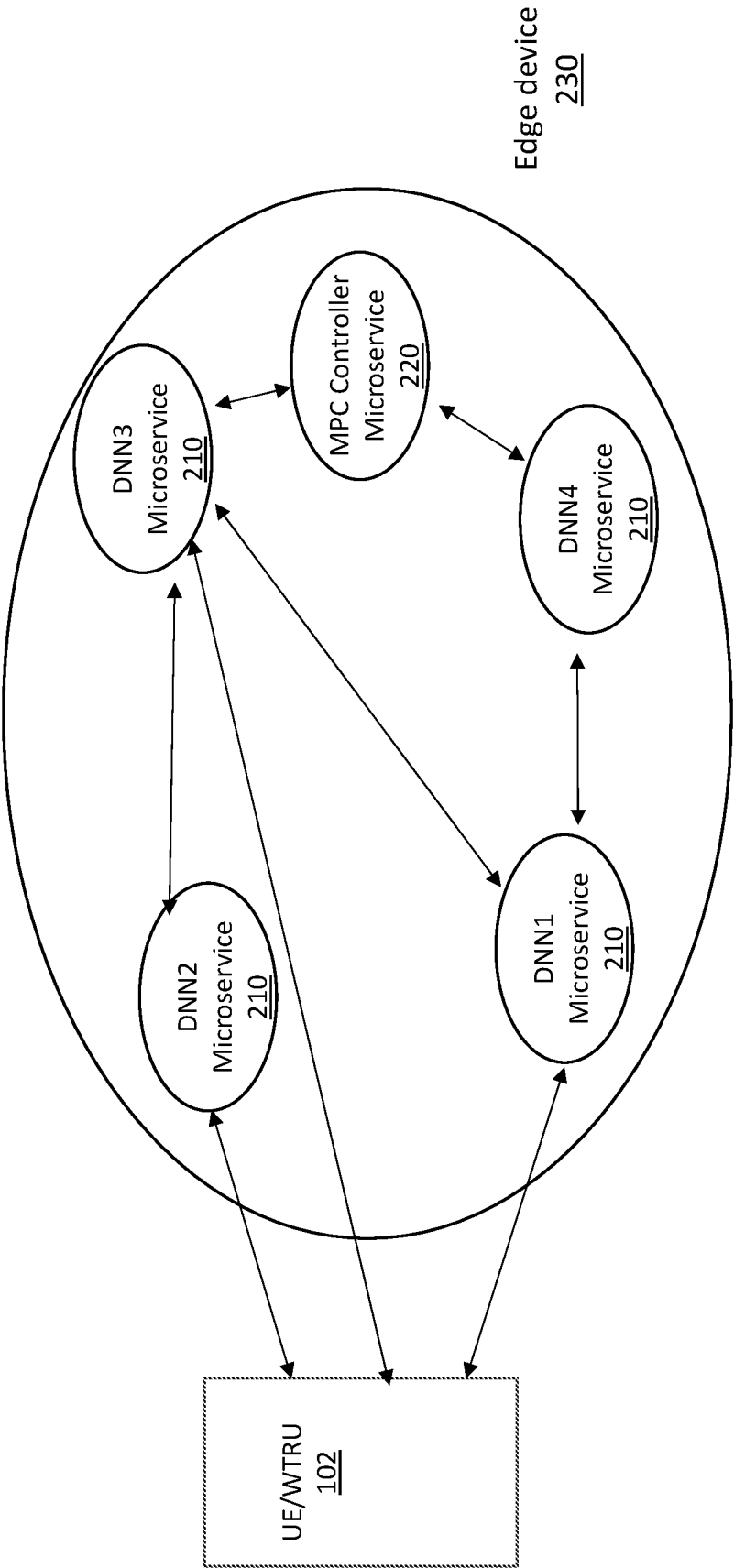


FIG. 2

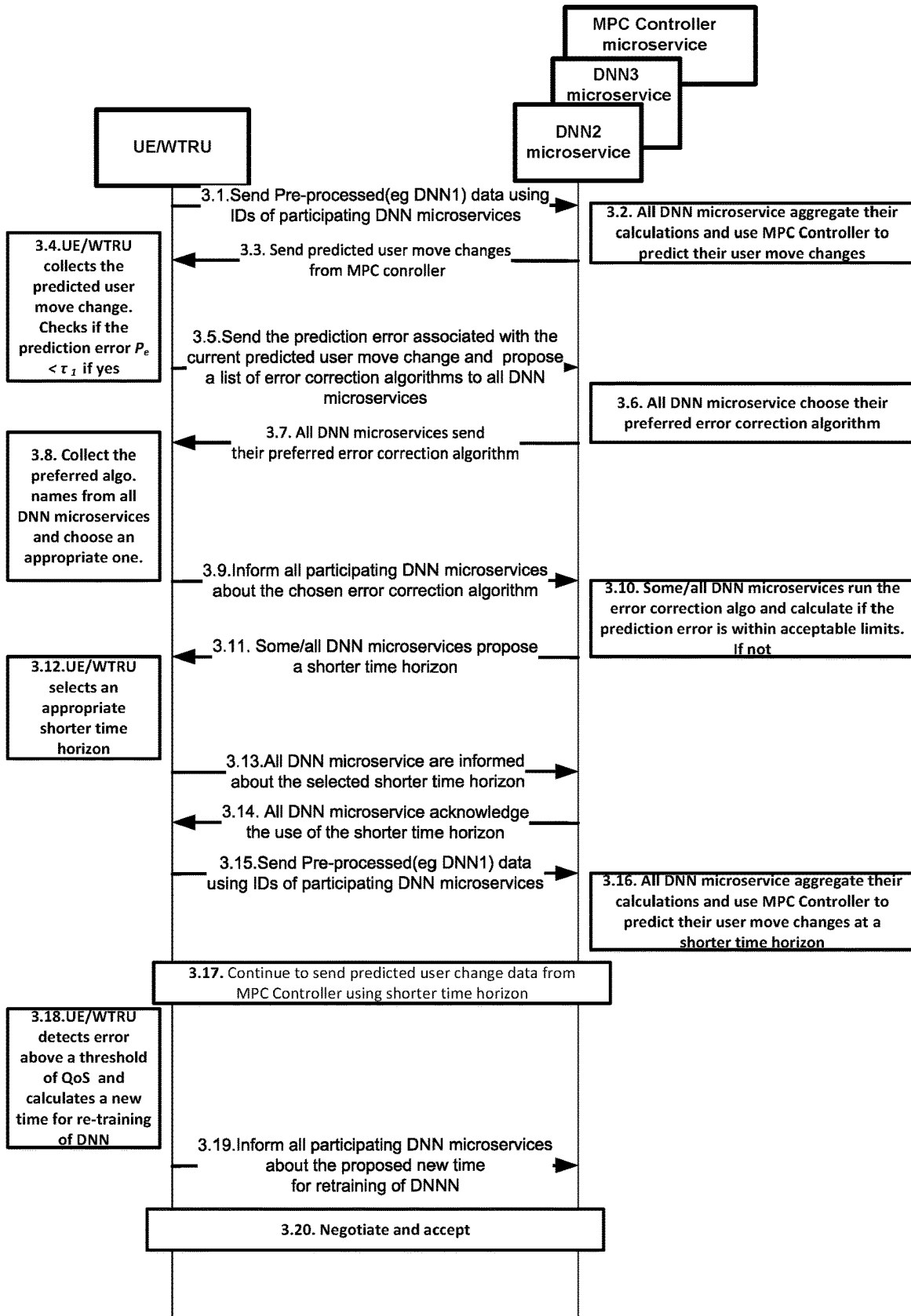


FIG. 3

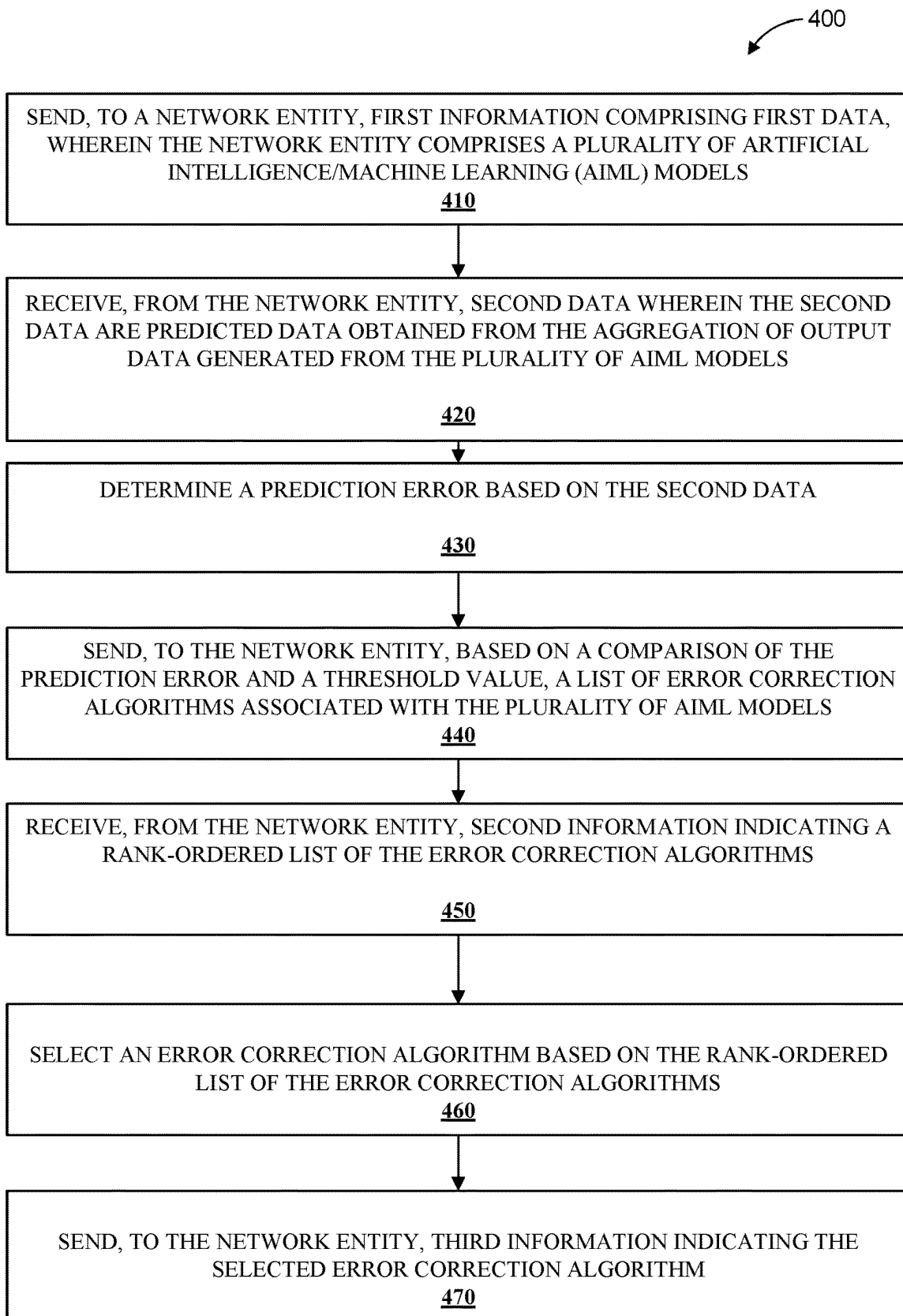


FIG. 4

**METHODS, ARCHITECTURES,
APPARATUSES AND SYSTEMS FOR
DATA-DRIVEN PREDICTION OF EXTENDED
REALITY (XR) DEVICE USER INPUTS BY
MULTIPLE DISTRIBUTED DEEP NEURAL
NETWORKS (DNNS)**

BACKGROUND

[0001] The present disclosure is generally directed to the fields of communications, software and encoding, including, for example, to methods, architectures, apparatuses, systems directed for data-driven prediction, for example for distributed XR enablers for a wireless transmit/receive unit (WTRU), and/or distributed Machine learning for XR applications.

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] A more detailed understanding may be had from the detailed description below, given by way of example in conjunction with drawings appended hereto. Figures in such drawings, like the detailed description, are examples. As such, the Figures (FIGs.) and the detailed description are not to be considered limiting, and other equally effective examples are possible and likely. Furthermore, like reference numerals (“ref.”) in the FIGs. indicate like elements, and wherein:

[0003] FIG. 1A is a system diagram illustrating an example communications system;

[0004] FIG. 1B is a system diagram illustrating an example wireless transmit/receive unit (WTRU) that may be used within the communications system illustrated in FIG. 1A;

[0005] FIG. 1C is a system diagram illustrating an example radio access network (RAN) and an example core network (CN) that may be used within the communications system illustrated in FIG. 1A;

[0006] FIG. 1D is a system diagram illustrating a further example RAN and a further example CN that may be used within the communications system illustrated in FIG. 1A;

[0007] FIG. 2 depicts a distributed overlay network formed by multiple deep neural networks (DNNS) and a model-predictive-control (MPC) controller that are running on an edge cloud as microservices;

[0008] FIG. 3 is a message sequence diagram illustrating messages exchanged between a WTRU (e.g., a UE) and multiple DNN microservices and a controller microservice; and

[0009] FIG. 4 is a diagram illustrating a method implemented by a WTRU for retraining multiple DNNS running on an edge cloud as microservices.

DETAILED DESCRIPTION

[0010] In the following detailed description, numerous specific details are set forth to provide a thorough understanding of embodiments and/or examples disclosed herein. However, it will be understood that such embodiments and examples may be practiced without some or all of the specific details set forth herein. In other instances, well-known methods, procedures, components and circuits have not been described in detail, so as not to obscure the following description. Further, embodiments and examples not specifically described herein may be practiced in lieu of, or in combination with, the embodiments and other

examples described, disclosed or otherwise provided explicitly, implicitly and/or inherently (collectively “provided”) herein. Although various embodiments are described and/or claimed herein in which an apparatus, system, device, etc. and/or any element thereof carries out an operation, process, algorithm, function, etc. and/or any portion thereof, it is to be understood that any embodiments described and/or claimed herein assume that any apparatus, system, device, etc. and/or any element thereof is configured to carry out any operation, process, algorithm, function, etc. and/or any portion thereof.

Example Communications System

[0011] The methods, apparatuses and systems provided herein are well-suited for communications involving both wired and wireless networks. An overview of various types of wireless devices and infrastructure is provided with respect to FIGS. 1A-1D, where various elements of the network may utilize, perform, be arranged in accordance with and/or be adapted and/or configured for the methods, apparatuses and systems provided herein.

[0012] FIG. 1A is a system diagram illustrating an example communications system 100 in which one or more disclosed embodiments may be implemented. The communications system 100 may be a multiple access system that provides content, such as voice, data, video, messaging, broadcast, etc., to multiple wireless users. The communications system 100 may enable multiple wireless users to access such content through the sharing of system resources, including wireless bandwidth. For example, the communications systems 100 may employ one or more channel access methods, such as code division multiple access (CDMA), time division multiple access (TDMA), frequency division multiple access (FDMA), orthogonal FDMA (OFDMA), single-carrier FDMA (SC-FDMA), zero-tail (ZT) unique-word (UW) discrete Fourier transform (DFT) spread OFDM (ZT UW DTS-s OFDM), unique word OFDM (UW-OFDM), resource block-filtered OFDM, filter bank multicarrier (FBMC), and the like.

[0013] As shown in FIG. 1A, the communications system 100 may include wireless transmit/receive units (WTRUs) 102a, 102b, 102c, 102d, a radio access network (RAN) 104/113, a core network (CN) 106/115, a public switched telephone network (PSTN) 108, the Internet 110, and other networks 112, though it will be appreciated that the disclosed embodiments contemplate any number of WTRUs, base stations, networks, and/or network elements. Each of the WTRUs 102a, 102b, 102c, 102d may be any type of device configured to operate and/or communicate in a wireless environment. By way of example, the WTRUs 102a, 102b, 102c, 102d, any of which may be referred to as a “station” and/or a “STA”, may be configured to transmit and/or receive wireless signals and may include (or be) a user equipment (UE), a mobile station, a fixed or mobile subscriber unit, a subscription-based unit, a pager, a cellular telephone, a personal digital assistant (PDA), a smartphone, a laptop, a netbook, a personal computer, a wireless sensor, a hotspot or Mi-Fi device, an Internet of Things (IoT) device, a watch or other wearable, a head-mounted display (HMD), a vehicle, a drone, a medical device and applications (e.g., remote surgery), an industrial device and applications (e.g., a robot and/or other wireless devices operating in an industrial and/or an automated processing chain contexts), a consumer electronics device, a device operating on

commercial and/or industrial wireless networks, and the like. Any of the WTRUs 102a, 102b, 102c and 102d may be interchangeably referred to as a UE.

[0014] The communications systems 100 may also include a base station 114a and/or a base station 114b. Each of the base stations 114a, 114b may be any type of device configured to wirelessly interface with at least one of the WTRUs 102a, 102b, 102c, 102d, e.g., to facilitate access to one or more communication networks, such as the CN 106/115, the Internet 110, and/or the networks 112. By way of example, the base stations 114a, 114b may be any of a base transceiver station (BTS), a Node-B (NB), an eNode-B (eNB), a Home Node-B (HNB), a Home eNode-B (HeNB), a gNode-B (gNB), a NR Node-B (NR NB), a site controller, an access point (AP), a wireless router, and the like. While the base stations 114a, 114b are each depicted as a single element, it will be appreciated that the base stations 114a, 114b may include any number of interconnected base stations and/or network elements.

[0015] The base station 114a may be part of the RAN 104/113, which may also include other base stations and/or network elements (not shown), such as a base station controller (BSC), a radio network controller (RNC), relay nodes, etc. The base station 114a and/or the base station 114b may be configured to transmit and/or receive wireless signals on one or more carrier frequencies, which may be referred to as a cell (not shown). These frequencies may be in licensed spectrum, unlicensed spectrum, or a combination of licensed and unlicensed spectrum. A cell may provide coverage for a wireless service to a specific geographical area that may be relatively fixed or that may change over time. The cell may further be divided into cell sectors. For example, the cell associated with the base station 114a may be divided into three sectors. Thus, in an embodiment, the base station 114a may include three transceivers, i.e., one for each sector of the cell. In an embodiment, the base station 114a may employ multiple-input multiple output (MIMO) technology and may utilize multiple transceivers for each or any sector of the cell. For example, beamforming may be used to transmit and/or receive signals in desired spatial directions.

[0016] The base stations 114a, 114b may communicate with one or more of the WTRUs 102a, 102b, 102c, 102d over an air interface 116, which may be any suitable wireless communication link (e.g., radio frequency (RF), microwave, centimeter wave, millimeter wave, infrared (IR), ultraviolet (UV), visible light, etc.). The air interface 116 may be established using any suitable radio access technology (RAT).

[0017] More specifically, as noted above, the communications system 100 may be a multiple access system and may employ one or more channel access schemes, such as CDMA, TDMA, FDMA, OFDMA, SC-FDMA, and the like. For example, the base station 114a in the RAN 104/113 and the WTRUs 102a, 102b, 102c may implement a radio technology such as Universal Mobile Telecommunications System (UMTS) Terrestrial Radio Access (UTRA), which may establish the air interface 116 using wideband CDMA (WCDMA). WCDMA may include communication protocols such as High-Speed Packet Access (HSPA) and/or Evolved HSPA (HSPA+). HSPA may include High-Speed Downlink Packet Access (HSDPA) and/or High-Speed Uplink Packet Access (HSUPA).

[0018] In an embodiment, the base station 114a and the WTRUs 102a, 102b, 102c may implement a radio technology such as Evolved UMTS Terrestrial Radio Access (E-UTRA), which may establish the air interface 116 using Long Term Evolution (LTE) and/or LTE-Advanced (LTE-A) and/or LTE-Advanced Pro (LTE-A Pro).

[0019] In an embodiment, the base station 114a and the WTRUs 102a, 102b, 102c may implement a radio technology such as NR Radio Access, which may establish the air interface 116 using New Radio (NR).

[0020] In an embodiment, the base station 114a and the WTRUs 102a, 102b, 102c may implement multiple radio access technologies. For example, the base station 114a and the WTRUs 102a, 102b, 102c may implement LTE radio access and NR radio access together, for instance using dual connectivity (DC) principles. Thus, the air interface utilized by WTRUs 102a, 102b, 102c may be characterized by multiple types of radio access technologies and/or transmissions sent to/from multiple types of base stations (e.g., an eNB and a gNB).

[0021] In an embodiment, the base station 114a and the WTRUs 102a, 102b, 102c may implement radio technologies such as IEEE 802.11 (i.e., Wireless Fidelity (Wi-Fi)), IEEE 802.16 (i.e., Worldwide Interoperability for Microwave Access (WiMAX)), CDMA2000, CDMA2000 1X, CDMA2000 EV-DO, Interim Standard 2000 (IS-2000), Interim Standard 95 (IS-95), Interim Standard 856 (IS-856), Global System for Mobile communications (GSM), Enhanced Data rates for GSM Evolution (EDGE), GSM EDGE (GERAN), and the like.

[0022] The base station 114b in FIG. 1A may be a wireless router, Home Node-B, Home eNode-B, or access point, for example, and may utilize any suitable RAT for facilitating wireless connectivity in a localized area, such as a place of business, a home, a vehicle, a campus, an industrial facility, an air corridor (e.g., for use by drones), a roadway, and the like. In an embodiment, the base station 114b and the WTRUs 102c, 102d may implement a radio technology such as IEEE 802.11 to establish a wireless local area network (WLAN). In an embodiment, the base station 114b and the WTRUs 102c, 102d may implement a radio technology such as IEEE 802.15 to establish a wireless personal area network (WPAN). In an embodiment, the base station 114b and the WTRUs 102c, 102d may utilize a cellular-based RAT (e.g., WCDMA, CDMA2000, GSM, LTE, LTE-A, LTE-A Pro, NR, etc.) to establish any of a small cell, picocell or femtocell. As shown in FIG. 1A, the base station 114b may have a direct connection to the Internet 110. Thus, the base station 114b may not be required to access the Internet 110 via the CN 106/115.

[0023] The RAN 104/113 may be in communication with the CN 106/115, which may be any type of network configured to provide voice, data, applications, and/or voice over internet protocol (VOIP) services to one or more of the WTRUs 102a, 102b, 102c, 102d. The data may have varying quality of service (QoS) requirements, such as differing throughput requirements, latency requirements, error tolerance requirements, reliability requirements, data throughput requirements, mobility requirements, and the like. The CN 106/115 may provide call control, billing services, mobile location-based services, pre-paid calling, Internet connectivity, video distribution, etc., and/or perform high-level security functions, such as user authentication. Although not shown in FIG. 1A, it will be appreciated that the RAN

104/113 and/or the **CN 106/115** may be in direct or indirect communication with other RANs that employ the same RAT as the **RAN 104/113** or a different RAT. For example, in addition to being connected to the **RAN 104/113**, which may be utilizing an NR radio technology, the **CN 106/115** may also be in communication with another RAN (not shown) employing any of a GSM, UMTS, CDMA 2000, WiMAX, E-UTRA, or Wi-Fi radio technology.

[0024] The **CN 106/115** may also serve as a gateway for the **WTRUs 102a, 102b, 102c, 102d** to access the **PSTN 108**, the **Internet 110**, and/or other networks **112**. The **PSTN 108** may include circuit-switched telephone networks that provide plain old telephone service (POTS). The **Internet 110** may include a global system of interconnected computer networks and devices that use common communication protocols, such as the transmission control protocol (TCP), user datagram protocol (UDP) and/or the internet protocol (IP) in the TCP/IP internet protocol suite. The networks **112** may include wired and/or wireless communications networks owned and/or operated by other service providers. For example, the networks **112** may include another CN connected to one or more RANs, which may employ the same RAT as the **RAN 104/114** or a different RAT.

[0025] Some or all of the **WTRUs 102a, 102b, 102c, 102d** in the communications system **100** may include multi-mode capabilities (e.g., the **WTRUs 102a, 102b, 102c, 102d** may include multiple transceivers for communicating with different wireless networks over different wireless links). For example, the **WTRU 102c** shown in FIG. 1A may be configured to communicate with the base station **114a**, which may employ a cellular-based radio technology, and with the base station **114b**, which may employ an IEEE 802 radio technology.

[0026] FIG. 1B is a system diagram illustrating an example **WTRU 102**. As shown in FIG. 1B, the **WTRU 102** may include a processor **118**, a transceiver **120**, a transmit/receive element **122**, a speaker/microphone **124**, a keypad **126**, a display/touchpad **128**, non-removable memory **130**, removable memory **132**, a power source **134**, a global positioning system (GPS) chipset **136**, and/or other elements/peripherals **138**, among others. It will be appreciated that the **WTRU 102** may include any sub-combination of the foregoing elements while remaining consistent with an embodiment.

[0027] The processor **118** may be a general purpose processor, a special purpose processor, a conventional processor, a digital signal processor (DSP), a plurality of microprocessors, one or more microprocessors in association with a DSP core, a controller, a microcontroller, Application Specific Integrated Circuits (ASICs), Field Programmable Gate Arrays (FPGAs) circuits, any other type of integrated circuit (IC), a state machine, and the like. The processor **118** may perform signal coding, data processing, power control, input/output processing, and/or any other functionality that enables the **WTRU 102** to operate in a wireless environment. The processor **118** may be coupled to the transceiver **120**, which may be coupled to the transmit/receive element **122**. While FIG. 1B depicts the processor **118** and the transceiver **120** as separate components, it will be appreciated that the processor **118** and the transceiver **120** may be integrated together, e.g., in an electronic package or chip.

[0028] The transmit/receive element **122** may be configured to transmit signals to, or receive signals from, a base station (e.g., the base station **114a**) over the air interface **116**.

For example, in an embodiment, the transmit/receive element **122** may be an antenna configured to transmit and/or receive RF signals. In an embodiment, the transmit/receive element **122** may be an emitter/detector configured to transmit and/or receive IR, UV, or visible light signals, for example. In an embodiment, the transmit/receive element **122** may be configured to transmit and/or receive both RF and light signals. It will be appreciated that the transmit/receive element **122** may be configured to transmit and/or receive any combination of wireless signals.

[0029] Although the transmit/receive element **122** is depicted in FIG. 1B as a single element, the **WTRU 102** may include any number of transmit/receive elements **122**. For example, the **WTRU 102** may employ MIMO technology. Thus, in an embodiment, the **WTRU 102** may include two or more transmit/receive elements **122** (e.g., multiple antennas) for transmitting and receiving wireless signals over the air interface **116**.

[0030] The transceiver **120** may be configured to modulate the signals that are to be transmitted by the transmit/receive element **122** and to demodulate the signals that are received by the transmit/receive element **122**. As noted above, the **WTRU 102** may have multi-mode capabilities. Thus, the transceiver **120** may include multiple transceivers for enabling the **WTRU 102** to communicate via multiple RATs, such as NR and IEEE 802.11, for example.

[0031] The processor **118** of the **WTRU 102** may be coupled to, and may receive user input data from, the speaker/microphone **124**, the keypad **126**, and/or the display/touchpad **128** (e.g., a liquid crystal display (LCD) display unit or organic light-emitting diode (OLED) display unit). The processor **118** may also output user data to the speaker/microphone **124**, the keypad **126**, and/or the display/touchpad **128**. In addition, the processor **118** may access information from, and store data in, any type of suitable memory, such as the non-removable memory **130** and/or the removable memory **132**. The non-removable memory **130** may include random-access memory (RAM), read-only memory (ROM), a hard disk, or any other type of memory storage device. The removable memory **132** may include a subscriber identity module (SIM) card, a memory stick, a secure digital (SD) memory card, and the like. In other embodiments, the processor **118** may access information from, and store data in, memory that is not physically located on the **WTRU 102**, such as on a server or a home computer (not shown).

[0032] The processor **118** may receive power from the power source **134**, and may be configured to distribute and/or control the power to the other components in the **WTRU 102**. The power source **134** may be any suitable device for powering the **WTRU 102**. For example, the power source **134** may include one or more dry cell batteries (e.g., nickel-cadmium (NiCd), nickel-zinc (NiZn), nickel metal hydride (NiMH), lithium-ion (Li-ion), etc.), solar cells, fuel cells, and the like.

[0033] The processor **118** may also be coupled to the GPS chipset **136**, which may be configured to provide location information (e.g., longitude and latitude) regarding the current location of the **WTRU 102**. In addition to, or in lieu of, the information from the GPS chipset **136**, the **WTRU 102** may receive location information over the air interface **116** from a base station (e.g., base stations **114a, 114b**) and/or determine its location based on the timing of the signals being received from two or more nearby base stations. It will

be appreciated that the WTRU 102 may acquire location information by way of any suitable location-determination method while remaining consistent with an embodiment.

[0034] The processor 118 may further be coupled to other elements/peripherals 138, which may include one or more software and/or hardware modules/units that provide additional features, functionality and/or wired or wireless connectivity. For example, the elements/peripherals 138 may include an accelerometer, an e-compass, a satellite transceiver, a digital camera (e.g., for photographs and/or video), a universal serial bus (USB) port, a vibration device, a television transceiver, a hands free headset, a Bluetooth® module, a frequency modulated (FM) radio unit, a digital music player, a media player, a video game player module, an Internet browser, a virtual reality and/or augmented reality (VR/AR) device, an activity tracker, and the like. The elements/peripherals 138 may include one or more sensors, the sensors may be one or more of a gyroscope, an accelerometer, a hall effect sensor, a magnetometer, an orientation sensor, a proximity sensor, a temperature sensor, a time sensor, a geolocation sensor, an altimeter, a light sensor, a touch sensor, a magnetometer, a barometer, a gesture sensor, a biometric sensor, and/or a humidity sensor.

[0035] The WTRU 102 may include a full duplex radio for which transmission and reception of some or all of the signals (e.g., associated with particular subframes for both the uplink (e.g., for transmission) and downlink (e.g., for reception) may be concurrent and/or simultaneous. The full duplex radio may include an interference management unit to reduce and/or substantially eliminate self-interference via either hardware (e.g., a choke) or signal processing via a processor (e.g., a separate processor (not shown) or via processor 118). In an embodiment, the WTRU 102 may include a half-duplex radio for which transmission and reception of some or all of the signals (e.g., associated with particular subframes for either the uplink (e.g., for transmission) or the downlink (e.g., for reception)).

[0036] FIG. 1C is a system diagram illustrating the RAN 104 and the CN 106 according to an embodiment. As noted above, the RAN 104 may employ an E-UTRA radio technology to communicate with the WTRUs 102a, 102b, and 102c over the air interface 116. The RAN 104 may also be in communication with the CN 106.

[0037] The RAN 104 may include eNode-Bs 160a, 160b, 160c, though it will be appreciated that the RAN 104 may include any number of eNode-Bs while remaining consistent with an embodiment. The eNode-Bs 160a, 160b, 160c may each include one or more transceivers for communicating with the WTRUs 102a, 102b, 102c over the air interface 116. In an embodiment, the eNode-Bs 160a, 160b, 160c may implement MIMO technology. Thus, the eNode-B 160a, for example, may use multiple antennas to transmit wireless signals to, and receive wireless signals from, the WTRU 102a.

[0038] Each of the eNode-Bs 160a, 160b, and 160c may be associated with a particular cell (not shown) and may be configured to handle radio resource management decisions, handover decisions, scheduling of users in the uplink (UL) and/or downlink (DL), and the like. As shown in FIG. 1C, the eNode-Bs 160a, 160b, 160c may communicate with one another over an X2 interface.

[0039] The CN 106 shown in FIG. 1C may include a mobility management entity (MME) 162, a serving gateway (SGW) 164, and a packet data network (PDN) gateway

(PGW) 166. While each of the foregoing elements are depicted as part of the CN 106, it will be appreciated that any one of these elements may be owned and/or operated by an entity other than the CN operator.

[0040] The MME 162 may be connected to each of the eNode-Bs 160a, 160b, and 160c in the RAN 104 via an SI interface and may serve as a control node. For example, the MME 162 may be responsible for authenticating users of the WTRUs 102a, 102b, 102c, bearer activation/deactivation, selecting a particular serving gateway during an initial attach of the WTRUs 102a, 102b, 102c, and the like. The MME 162 may provide a control plane function for switching between the RAN 104 and other RANs (not shown) that employ other radio technologies, such as GSM and/or WCDMA.

[0041] The SGW 164 may be connected to each of the eNode-Bs 160a, 160b, 160c in the RAN 104 via the SI interface. The SGW 164 may generally route and forward user data packets to/from the WTRUs 102a, 102b, 102c. The SGW 164 may perform other functions, such as anchoring user planes during inter-eNode-B handovers, triggering paging when DL data is available for the WTRUs 102a, 102b, 102c, managing and storing contexts of the WTRUs 102a, 102b, 102c, and the like.

[0042] The SGW 164 may be connected to the PGW 166, which may provide the WTRUs 102a, 102b, 102c with access to packet-switched networks, such as the Internet 110, to facilitate communications between the WTRUs 102a, 102b, 102c and IP-enabled devices.

[0043] The CN 106 may facilitate communications with other networks. For example, the CN 106 may provide the WTRUs 102a, 102b, 102c with access to circuit-switched networks, such as the PSTN 108, to facilitate communications between the WTRUs 102a, 102b, 102c and traditional land-line communications devices. For example, the CN 106 may include, or may communicate with, an IP gateway (e.g., an IP multimedia subsystem (IMS) server) that serves as an interface between the CN 106 and the PSTN 108. In addition, the CN 106 may provide the WTRUs 102a, 102b, 102c with access to the other networks 112, which may include other wired and/or wireless networks that are owned and/or operated by other service providers.

[0044] Although the WTRU is described in FIGS. 1A-1D as a wireless terminal, it is contemplated that in certain representative embodiments that such a terminal may use (e.g., temporarily or permanently) wired communication interfaces with the communication network. In representative embodiments, the other network 112 may be a WLAN.

[0045] A WLAN in infrastructure basic service set (BSS) mode may have an access point (AP) for the BSS and one or more stations (STAs) associated with the AP. The AP may have an access or an interface to a distribution system (DS) or another type of wired/wireless network that carries traffic into and/or out of the BSS. Traffic to STAs that originates from outside the BSS may arrive through the AP and may be delivered to the STAs. Traffic originating from STAs to destinations outside the BSS may be sent to the AP to be delivered to respective destinations. Traffic between STAs within the BSS may be sent through the AP, for example, where the source STA may send traffic to the AP and the AP may deliver the traffic to the destination STA. The traffic between STAs within a BSS may be considered and/or referred to as peer-to-peer traffic. The peer-to-peer traffic may be sent between (e.g., directly between) the source and

destination STAs with a direct link setup (DLS). In certain representative embodiments, the DLS may use an 802.11e DLS or an 802.11z tunneled DLS (TDLS). A WLAN using an Independent BSS (IBSS) mode may not have an AP, and the STAs (e.g., all of the STAs) within or using the IBSS may communicate directly with each other. The IBSS mode of communication may sometimes be referred to herein as an “ad-hoc” mode of communication.

[0046] When using the 802.11ac infrastructure mode of operation or a similar mode of operations, the AP may transmit a beacon on a fixed channel, such as a primary channel. The primary channel may be a fixed width (e.g., 20 MHz wide bandwidth) or a dynamically set width via signaling. The primary channel may be the operating channel of the BSS and may be used by the STAs to establish a connection with the AP. In certain representative embodiments, Carrier sense multiple access with collision avoidance (CSMA/CA) may be implemented, for example in in 802.11 systems. For CSMA/CA, the STAs (e.g., every STA), including the AP, may sense the primary channel. If the primary channel is sensed/detected and/or determined to be busy by a particular STA, the particular STA may back off. One STA (e.g., only one station) may transmit at any given time in a given BSS.

[0047] High throughput (HT) STAs may use a 40 MHz wide channel for communication, for example, via a combination of the primary 20 MHz channel with an adjacent or nonadjacent 20 MHz channel to form a 40 MHz wide channel.

[0048] Very high throughput (VHT) STAs may support 20 MHz, 40 MHz, 80 MHz, and/or 160 MHz wide channels. The 40 MHz, and/or 80 MHz, channels may be formed by combining contiguous 20 MHz channels. A 160 MHz channel may be formed by combining 8 contiguous 20 MHz channels, or by combining two non-contiguous 80 MHz channels, which may be referred to as an 80+80 configuration. For the 80+80 configuration, the data, after channel encoding, may be passed through a segment parser that may divide the data into two streams. Inverse fast Fourier transform (IFFT) processing, and time domain processing, may be done on each stream separately. The streams may be mapped on to the two 80 MHz channels, and the data may be transmitted by a transmitting STA. At the receiver of the receiving STA, the above-described operation for the 80+80 configuration may be reversed, and the combined data may be sent to a medium access control (MAC) layer, entity, etc.

[0049] Sub 1 GHz modes of operation are supported by 802.11af and 802.11ah. The channel operating bandwidths, and carriers, are reduced in 802.11af and 802.11ah relative to those used in 802.11n, and 802.11ac. 802.11af supports 5 MHz, 10 MHz and 20 MHz bandwidths in the TV white space (TVWS) spectrum, and 802.11ah supports 1 MHz, 2 MHz, 4 MHz, 8 MHz, and 16 MHz bandwidths using non-TVWS spectrum. According to a representative embodiment, 802.11ah may support meter type control/machine-type communications (MTC), such as MTC devices in a macro coverage area. MTC devices may have certain capabilities, for example, limited capabilities including support for (e.g., only support for) certain and/or limited bandwidths. The MTC devices may include a battery with a battery life above a threshold (e.g., to maintain a very long battery life).

[0050] WLAN systems, which may support multiple channels, and channel bandwidths, such as 802.11n, 802.11ac,

802.11af, and 802.11ah, include a channel which may be designated as the primary channel. The primary channel may have a bandwidth equal to the largest common operating bandwidth supported by all STAs in the BSS. The bandwidth of the primary channel may be set and/or limited by a STA, from among all STAs in operating in a BSS, which supports the smallest bandwidth operating mode. In the example of 802.11ah, the primary channel may be 1 MHz wide for STAs (e.g., MTC type devices) that support (e.g., only support) a 1 MHz mode, even if the AP, and other STAs in the BSS support 2 MHz, 4 MHz, 8 MHz, 16 MHz, and/or other channel bandwidth operating modes. Carrier sensing and/or network allocation vector (NAV) settings may depend on the status of the primary channel. If the primary channel is busy, for example, due to a STA (which supports only a 1 MHz operating mode), transmitting to the AP, the entire available frequency bands may be considered busy even though a majority of the frequency bands remains idle and may be available.

[0051] In the United States, the available frequency bands, which may be used by 802.11ah, are from 902 MHz to 928 MHz. In Korea, the available frequency bands are from 917.5 MHz to 923.5 MHz. In Japan, the available frequency bands are from 916.5 MHz to 927.5 MHz. The total bandwidth available for 802.11ah is 6 MHz to 26 MHz depending on the country code.

[0052] FIG. 1D is a system diagram illustrating the RAN 113 and the CN 115 according to an embodiment. As noted above, the RAN 113 may employ an NR radio technology to communicate with the WTRUs 102a, 102b, 102c over the air interface 116. The RAN 113 may also be in communication with the CN 115.

[0053] The RAN 113 may include gNBs 180a, 180b, 180c, though it will be appreciated that the RAN 113 may include any number of gNBs while remaining consistent with an embodiment. The gNBs 180a, 180b, 180c may each include one or more transceivers for communicating with the WTRUs 102a, 102b, 102c over the air interface 116. In an embodiment, the gNBs 180a, 180b, 180c may implement MIMO technology. For example, gNBs 180a, 180b may utilize beamforming to transmit signals to and/or receive signals from the WTRUs 102a, 102b, 102c. Thus, the gNB 180a, for example, may use multiple antennas to transmit wireless signals to, and/or receive wireless signals from, the WTRU 102a. In an embodiment, the gNBs 180a, 180b, 180c may implement carrier aggregation technology. For example, the gNB 180a may transmit multiple component carriers to the WTRU 102a (not shown). A subset of these component carriers may be on unlicensed spectrum while the remaining component carriers may be on licensed spectrum. In an embodiment, the gNBs 180a, 180b, 180c may implement Coordinated Multi-Point (COMP) technology. For example, WTRU 102a may receive coordinated transmissions from gNB 180a and gNB 180b (and/or gNB 180c).

[0054] The WTRUs 102a, 102b, 102c may communicate with gNBs 180a, 180b, 180c using transmissions associated with a scalable numerology. For example, OFDM symbol spacing and/or OFDM subcarrier spacing may vary for different transmissions, different cells, and/or different portions of the wireless transmission spectrum. The WTRUs 102a, 102b, 102c may communicate with gNBs 180a, 180b, 180c using subframe or transmission time intervals (TTIs) of

various or scalable lengths (e.g., including a varying number of OFDM symbols and/or lasting varying lengths of absolute time).

[0055] The gNBs 180a, 180b, 180c may be configured to communicate with the WTRUs 102a, 102b, 102c in a standalone configuration and/or a non-standalone configuration. In the standalone configuration, WTRUs 102a, 102b, 102c may communicate with gNBs 180a, 180b, 180c without also accessing other RANs (e.g., such as eNode-Bs 160a, 160b, 160c). In the standalone configuration, WTRUs 102a, 102b, 102c may utilize one or more of gNBs 180a, 180b, 180c as a mobility anchor point. In the standalone configuration, WTRUs 102a, 102b, 102c may communicate with gNBs 180a, 180b, 180c using signals in an unlicensed band. In a non-standalone configuration WTRUs 102a, 102b, 102c may communicate with/connect to gNBs 180a, 180b, 180c while also communicating with/connecting to another RAN such as eNode-Bs 160a, 160b, 160c. For example, WTRUs 102a, 102b, 102c may implement DC principles to communicate with one or more gNBs 180a, 180b, 180c and one or more eNode-Bs 160a, 160b, 160c substantially simultaneously. In the non-standalone configuration, eNode-Bs 160a, 160b, 160c may serve as a mobility anchor for WTRUs 102a, 102b, 102c and gNBs 180a, 180b, 180c may provide additional coverage and/or throughput for servicing WTRUs 102a, 102b, 102c.

[0056] Each of the gNBs 180a, 180b, 180c may be associated with a particular cell (not shown) and may be configured to handle radio resource management decisions, handover decisions, scheduling of users in the UL and/or DL, support of network slicing, dual connectivity, interworking between NR and E-UTRA, routing of user plane data towards user plane functions (UPFs) 184a, 184b, routing of control plane information towards access and mobility management functions (AMFs) 182a, 182b, and the like. As shown in FIG. 1D, the gNBs 180a, 180b, 180c may communicate with one another over an Xn interface.

[0057] The CN 115 shown in FIG. 1D may include at least one AMF 182a, 182b, at least one UPF 184a, 184b, at least one session management function (SMF) 183a, 183b, and at least one Data Network (DN) 185a, 185b. While each of the foregoing elements are depicted as part of the CN 115, it will be appreciated that any of these elements may be owned and/or operated by an entity other than the CN operator.

[0058] The AMF 182a, 182b may be connected to one or more of the gNBs 180a, 180b, 180c in the RAN 113 via an N2 interface and may serve as a control node. For example, the AMF 182a, 182b may be responsible for authenticating users of the WTRUs 102a, 102b, 102c, support for network slicing (e.g., handling of different protocol data unit (PDU) sessions with different requirements), selecting a particular SMF 183a, 183b, management of the registration area, termination of NAS signaling, mobility management, and the like. Network slicing may be used by the AMF 182a, 182b, e.g., to customize CN support for WTRUs 102a, 102b, 102c based on the types of services being utilized WTRUs 102a, 102b, 102c. For example, different network slices may be established for different use cases such as services relying on ultra-reliable low latency (URLLC) access, services relying on enhanced massive mobile broadband (eMBB) access, services for MTC access, and/or the like. The AMF 162 may provide a control plane function for switching between the RAN 113 and other RANs (not shown) that

employ other radio technologies, such as LTE, LTE-A, LTE-A Pro, and/or non-3GPP access technologies such as Wi-Fi.

[0059] The SMF 183a, 183b may be connected to an AMF 182a, 182b in the CN 115 via an N11 interface. The SMF 183a, 183b may also be connected to a UPF 184a, 184b in the CN 115 via an N4 interface. The SMF 183a, 183b may select and control the UPF 184a, 184b and configure the routing of traffic through the UPF 184a, 184b. The SMF 183a, 183b may perform other functions, such as managing and allocating UE IP address, managing PDU sessions, controlling policy enforcement and QoS, providing downlink data notifications, and the like. A PDU session type may be IP-based, non-IP based, Ethernet-based, and the like.

[0060] The UPF 184a, 184b may be connected to one or more of the gNBs 180a, 180b, 180c in the RAN 113 via an N3 interface, which may provide the WTRUs 102a, 102b, 102c with access to packet-switched networks, such as the Internet 110, e.g., to facilitate communications between the WTRUs 102a, 102b, 102c and IP-enabled devices. The UPF 184a, 184b may perform other functions, such as routing and forwarding packets, enforcing user plane policies, supporting multi-homed PDU sessions, handling user plane QoS, buffering downlink packets, providing mobility anchoring, and the like.

[0061] The CN 115 may facilitate communications with other networks. For example, the CN 115 may include, or may communicate with, an IP gateway (e.g., an IP multimedia subsystem (IMS) server) that serves as an interface between the CN 115 and the PSTN 108. In addition, the CN 115 may provide the WTRUs 102a, 102b, 102c with access to the other networks 112, which may include other wired and/or wireless networks that are owned and/or operated by other service providers. In an embodiment, the WTRUs 102a, 102b, 102c may be connected to a local Data Network (DN) 185a, 185b through the UPF 184a, 184b via the N3 interface to the UPF 184a, 184b and an N6 interface between the UPF 184a, 184b and the DN 185a, 185b.

[0062] In view of FIGS. 1A-1D, and the corresponding description of FIGS. 1A-1D, one or more, or all, of the functions described herein with regard to any of: WTRUs 102a-d, base stations 114a-b, eNode-Bs 160a-c, MME 162, SGW 164, PGW 166, gNBs 180a-c, AMFs 182a-b, UPFs 184a-b, SMFs 183a-b, DNs 185a-b, and/or any other element(s)/device(s) described herein, may be performed by one or more emulation elements/devices (not shown). The emulation devices may be one or more devices configured to emulate one or more, or all, of the functions described herein. For example, the emulation devices may be used to test other devices and/or to simulate network and/or WTRU functions.

[0063] The emulation devices may be designed to implement one or more tests of other devices in a lab environment and/or in an operator network environment. For example, the one or more emulation devices may perform the one or more, or all, functions while being fully or partially implemented and/or deployed as part of a wired and/or wireless communication network in order to test other devices within the communication network. The one or more emulation devices may perform the one or more, or all, functions while being temporarily implemented/deployed as part of a wired and/or wireless communication network. The emulation device may be directly coupled to another device for pur-

poses of testing and/or may performing testing using over-the-air wireless communications.

[0064] The one or more emulation devices may perform the one or more, including all, functions while not being implemented/deployed as part of a wired and/or wireless communication network. For example, the emulation devices may be utilized in a testing scenario in a testing laboratory and/or a non-deployed (e.g., testing) wired and/or wireless communication network in order to implement testing of one or more components. The one or more emulation devices may be test equipment. Direct RF coupling and/or wireless communications via RF circuitry (e.g., which may include one or more antennas) may be used by the emulation devices to transmit and/or receive data.

[0065] A scenario can be presented where a single artificial intelligence (AI)/machine learning (ML) model such as a deep neural network (DNN) for example is running on the WTRU (e.g., UE) and a single AI/ML model such as a DNN is running on the edge (see for example, reference [1]). In such a scenario, using a unique AI model in the WTRU (e.g., UE)/edge nodes may not provide the better KPI, for example because of heavy-tailed WTRU (e.g., UE) input distribution (the inputs and outputs to this model are discussed, for example, in detail in reference [1]).

[0066] A solution can be generalized (see for example, reference [1]) to a scenario where one AI/ML model such as a DNN is running on the WTRU (e.g., UE) and multiple AI/ML models such as a DNNs are running on the edge. This situation is illustrated in FIG. 2.

[0067] As shown in FIG. 2, multiple DNNs **210** and a (e.g., model-predictive-control (MPC)) controller **220** are running on the Edge cloud **230**, for example, as microservices and may form an overlay network with the WTRU (e.g., UE) **102**. Microservices are a type of service oriented architectural pattern where each microservice is deployed in a container (e.g., Docker) and may use lightweight protocol such as REpresentational State Transfer (REST) or Google Remote Procedure Call (gRPC) for communication. The WTRU (e.g., UE) itself also hosts a small DNN not shown in FIG. 2. Note that FIG. 2 shows four DNNs running on the edge, there could be less or more DNNs running there (for more details of MPC and WTRU (e.g., UE) architecture, see for example, reference [1]).

[0068] An overlay network is a virtual network consisting of nodes and virtual links which sits on top of an underlying network (see for example, reference [2]). The overlay network represents a task dependency graph which captures the precedence constraints between various microservices and so is dependent on the implementation of the distribution of DNNs (see for example, references [3] and [4]).

[0069] The DNNs on the edge may have more computational resources that are available to them. If a more accurate prediction is required than the one facilitated by the small DNN on the WTRU (e.g., UE), its data can be sent for further processing to the DNNs on the edge for a higher quality prediction. As discussed above, a scenario where there was only one DNN on the edge was presented. It can be envisaged multiple DNNs on the edge that can be utilized for more accurate prediction by piping data through one or more of them. The WTRU (e.g., UE) runs an algorithm that may decide how many DNNs to include for a given prediction.

[0070] This scenario leads to the problem to communicate and coordinate the DNN on the WTRU (e.g., UE) with the

multiple DNNs on the edge, for example, for: (i) triggering the online (re) training of models and for (ii) agreeing on the frequency of that training.

[0071] As presented above, it can be envisaged (see for example reference [1]) a solution for triggering the online (re) training of the models and for agreement on the frequency of that training in the case when there is a single DNN on the WTRU (e.g., UE) and one on the edge. However, when there are multiple DNNs on the edge, the following new problems need to be solved:

[0072] (1) The determination of the information flow between the DNN on the WTRU (e.g., UE) and the one on the Edge. In particular, the determination of the parameters used (e.g., required) to be exchanged between the DNN on the WTRU (e.g., UE) and the multiple DNNs on the Edge, for example, for the triggering and frequency of retraining of the models.

[0073] (2) The determination of the order of the messages that should be exchanged for a correct execution. With multiple DNNs running on the edge, a correct ordering of messages may be used (e.g., required) to coordinate them for triggering and deciding on the frequency of retraining of the models.

[0074] (3) The determination of the messages (e.g., need) to be exchanged to agree on the triggering and frequency of retraining of the models. In particular, the message sequences that may be used (e.g., needed) to ensure that (e.g., all) the DNNs agree on the same value of the time for triggering and the frequency of retraining of the model.

[0075] Methods and apparatus are provided to propose solution to the problem where an AI model distributed over a DNN running on the WTRU (e.g., UE) and multiple DNNs on the Edge that is used to predict future user actions gets out-of-date resulting in prediction errors are provided.

[0076] According to embodiments, a method and apparatus are provided to determine the parameters (e.g., required) to be exchanged between the DNN on the WTRU (e.g., UE) and the multiple DNNs on the Edge, for example, for the triggering and frequency of retraining of the models.

[0077] According to embodiments, a method and apparatus are provided to determine an (e.g., correct) ordering of messages that may be used (e.g., required) to coordinate the multiple DNNs, for example, for triggering and deciding on the frequency of retraining of the models.

[0078] According to embodiments, a method and apparatus are provided to determine the message sequences, for example, to ensure that all the DNNs agree on a value (e.g., the same value) of the time for triggering and the frequency of retraining of the model.

[0079] FIG. 3 is a message sequence diagram illustrating the resulting messages exchanged between the WTRU (e.g., UE) and multiple DNN microservices and a (e.g., MPC) controller microservice. Note that even though FIG. 3 shows two DNN microservices, the messages exchanged may be valid for any number of such microservices.

[0080] According to embodiments, the WTRU (e.g., UE) may send a message comprising (e.g., pre-processed) data, for example, from its DNN (labelled as DNN1 in FIG. 3) to one or more (e.g., all the) DNNs running as microservices on the Edge, for example using their IDs (e.g. fully qualified domain names). The WTRU (e.g., UE) may include its own id in the message. The data sent may comprise the early exit data that may be used by the multiple DNNs running on the

Edge as microservices for further processing and may be forwarded to the (e.g., MPC) controller microservice for prediction on the edge device (step 3.1). This prediction could be a point estimate or a probability distribution.

[0081] According to embodiments, one or more (e.g., all) DNN microservices may aggregate their outputs, for example, using their early exit data. This aggregated/combined data may be forwarded to the (e.g., MPC) controller microservice for prediction on the edge device (step 3.2). This prediction could be a point estimate or a probability distribution of the predicted user input.

[0082] According to embodiments, the (e.g., MPC) controller microservice running on the edge may send data to the WTRU (e.g., UE) using aggregated data from all DNNs (step 3.3). The data may comprise predicted user action (e.g., move) change, for example, in a game. This may include motion of the WTRU or the user picking up an item in the game, or the user changing their gaze to look sideways etc.

[0083] According to embodiments, the WTRU (e.g., UE) may determine/check the prediction error, for example, if the prediction error $P_e > \tau_1$ that is whether the error is above a threshold value (step 3.4). For example, the prediction error may be determined by comparing the prediction of the user action with the actual user action (e.g., in a game). If the error is above the threshold value, the WTRU (e.g., UE) may send (signal) a message to the Edge comprising a list of error correction algorithm running to all DNN microservices. Examples of these algorithms may include but are not limited to Kalman Filtering, Partial Cube rendering, Depth Map Rendering (see, for example reference [5]).

[0084] According to embodiments, the WTRU (e.g., UE) may send the prediction error associated with the current predicted user action (e.g., move) change and may propose a list of error correction algorithms to one or more (e.g., all) DNN microservices as discussed above (step 3.5).

[0085] According to embodiments, one or more (e.g., all) DNN microservices may rank-order their preferred error correction algorithm from the list they receive from the WTRU (e.g., UE) (step 3.6). This ranking may be done based on any of: the resources available to each microservice to run those algorithms, the code base availability of that algorithm or any other factor important to the operator of the microservice.

[0086] According to embodiments, one or more (e.g., all) DNN microservices may send their rank-ordered list of error correction algorithms back to the WTRU (e.g., UE) (step 3.7). This list could be for example (but not limited to) a set of integers or a set of strings encoding the name and/or parameters for those algorithms.

[0087] According to embodiments, the WTRU (e.g., UE) may receive information indicating (e.g., collect) the rank-ordered preferences of error-correction algorithm names from all the DNN microservices and chooses an appropriate one (step 3.8). For example, the WTRU (e.g., UE) could choose the algorithm ranked highest by a majority of DNN microservices.

[0088] According to embodiments, the WTRU (e.g., UE) may inform/signal one or more (e.g., all) participating DNN microservices about the chosen error correction algorithm so that they can calculate the error correction that may be used (e.g., necessary) for appropriate prediction fidelity/accuracy (step 3.9).

[0089] According to embodiments, one or more (e.g., all) DNN microservices running on the Edge device may calculate if after the running the error correction algorithms proposed by the WTRU (e.g., UE), the error is still above a threshold value (step 3.10). If that is the case, those microservices running on the Edge may send messages to the WTRU (e.g., UE), for example, comprising information indicating (e.g., suggesting) a shorter time horizon of prediction for a more accurate prediction.

[0090] According to embodiments, one or more (e.g., all) DNN microservices may send information indicating (e.g., propose) a shorter time horizon to the WTRU (e.g., UE) as discussed above (step 3.11).

[0091] According to embodiments, the WTRU (e.g., UE) may determine/select a (e.g., appropriate) shorter time horizon from the proposed time horizons sent by some/all of the DNN microservices (step 3.12). This selection could be done for example on the basis of the time horizon chosen by a maximum of microservices or the one which is the shortest amongst all proposals.

[0092] According to embodiments, the WTRU (e.g., UE) may inform/signal one or more (e.g., all) DNN microservices about the selected shorter time horizon that may be used for prediction, for example, of the user move (step 3.13). It can be assume/determine that the edge device that may host the DNN microservices may be in close network proximity of the WTRU (e.g., UE).

[0093] According to embodiments, one or more (e.g., all) DNN microservice may send information acknowledging the use of the shorter time horizon, for example, for the prediction of future user moves (step 3.14).

[0094] According to embodiments, the WTRU (e.g., UE) may send (e.g., pre-processed) data from its DNN (labelled as DNN1 in FIG. 3) to one or more (e.g., all) DNNs running as microservices on the Edge, for example, using their IDs (e.g., Fully Qualified Domain Names) (step 3.15). The data sent may be the early exit data calculated for a shorter time horizon that may be use by the multiple DNNs running on the Edge as microservices for further processing and may be forwarded to the (e.g., MPC) controller microservice for prediction on the edge device. This prediction could be a point estimate or a probability distribution.

[0095] According to embodiments, one or more (e.g., all) DNN microservice may aggregate their outputs, for example, using their early exit data calculated at a shorter time horizon (step 3.16). This aggregated/combined data may be forwarded to the (e.g., MPC) controller for prediction on the edge device. This prediction could be a point estimate or a probability distribution of the predicted user action (e.g., move) change (e.g., new user input).

[0096] According to embodiments, the (e.g., MPC) controller microservice running on the edge may (e.g., continue) to send to the WTRU (e.g., UE), predicted user move change at the shorter time horizon, for example, using aggregated data from one or more (e.g., all) DNNs, (step 3.17).

[0097] According to embodiments, the WTRU (e.g., UE) may observe/monitor the exchange of data to determine if the proposed shorter time horizon results in an error that, for example, may be within the QoS requirement of the user (step 3.18). If not, the WTRU (e.g., UE) may configure a (e.g., proposed) time for re-training the DNN on the WTRU (e.g., UE) and the microservices running on the edge.

[0098] According to embodiments, the WTRU (e.g., UE) may inform/signal one or more (e.g., all) participating DNN

microservices about the (e.g., proposed) new time for retraining of the DNN running on each microservice (step 3.19).

[0099] According to embodiments, the WTRU (e.g., UE) and the DNNs may negotiate and accept a new time for retraining of the DNN running on each microservice (step 3.20).

[0100] The following section proposes Extension to the Session Description Protocol (SDP) at the IETF.

[0101] The fields and data of our message exchanges of FIG. 3 can be mapped to SDP “Offer” procedure’s capability negotiation parameters (see for example references [6], [7]) contained in the UPDATE method of the session initiation protocol (SIP) (see for example reference [8]).

[0102] To add new capabilities to SDP, a new attribute type “a=” may be defined at the session level (see for example reference [9]). Attributes at the session level are listed before the first media line in SDP.

[0103] The following shows a set of proposed new SDP offer procedure’s capability attribute.

[0104] The novel message for step 3.1 can be sent as the following new SDP offer procedure’s capability attribute: a=dnn_data: dnn_id:value destination_dnn_id: id1 id2 id3 . . . idN dnn_param: parameter1 parameter2 . . . parameter. The parameter “a=dnn_data: ” may be the new attribute representing DNN data that has multiple parameters separated by space. The first parameter “dnn_id: ” may represent the label for the identifier of the origin DNN microservice and may have the parameter called “value”. The next parameter “destination_dnn_id” may represent the identifier of the destination DNN microservices and can have multiple identifier values. The parameter “dnn_param: ” attribute may represent DNN data that has multiple parameters separated by space.

[0105] The novel message for step 3.3 can be sent as the following new SDP offer procedure’s capability attribute: a=predict: mpc_id:value destination_dnn_id: id1 id2 id3 . . . idN predict_data: point_est: value prob_dist: parameter1 parameter2 . . . parameter. The parameter “a=predict: ” may be the new attribute representing the prediction that has multiple parameters separated by space. The first parameter “mpc_id: ” may represent the label for the identifier of the origin MPC microservice and has the parameter called “value”. The next parameter “destination_dnn_id” may represent the identifier of the destination DNN microservices and can have multiple identifier values. The parameter “predict_data” may represent the predicted value and has two parameters. The first parameter “point_est: ” may represent the label for a point estimate prediction and has the parameter called “value”. The next parameter “prob_dist: ” may represent the label for a probability distribution prediction and can have multiple parameters separated by space.

[0106] The novel message for step 3.5 can be sent as the following new SDP offer procedure’s capability attribute: a=predict_error: dnn_id:value destination_dnn_id: id1 id2 id3 . . . idN algo_list: parameter1 parameter 2 . . . parameterN error: parameter1 parameter2. parameterN. The parameter “a=predict_error: ” may be the new attribute representing prediction error that has multiple parameters separated by space. The first parameter “dnn_id: ” may represent the label for the identifier of the origin DNN microservice and has the parameter called “value”. The next parameter “destination_dnn_id” may represent the identifier

of the destination DNN microservices and can have multiple identifier values. The parameter “algo_list” may represent the label for the rank ordered list of algorithms followed by multiple parameters. The parameter “error” may represent prediction error that has multiple parameters separated by space.

[0107] The novel message for step 3.7 can be sent as the following new SDP offer procedure’s capability attribute: a=preferred_algo_list: dnn_id:value destination_dnn_id: id1 id2 id3 . . . idN rank_ordered_algo_list: parameter 1 parameter 2 . . . parameterN. The parameter “a=preferred_algo_list: ” may be the new attribute representing a rank ordered list of preferred algorithms of a DNN microservice that has multiple parameters separated by space. The first parameter “dnn_id: ” may represent the label for the identifier of the origin DNN microservice and has the parameter called “value”. The next parameter “destination_dnn_id” may represent the identifier of the destination DNN microservices and can have multiple identifier values. Finally, the parameter “rank_ordered_algo_list” may represent the identifier for the list of preferred algorithms followed by multiple parameters encoding the names and/or parameters of those algorithms.

[0108] The novel message for step 3.9 can be sent as the following new SDP offer procedure’s capability attribute: a=selected_algo: dnn_id:value destination_dnn_id: id1 id2 id3 . . . idN algo: parameter 1 parameter2 . . . parameterN. The parameter “a=selected_algo: ” may be the new attribute representing the algorithm selected by the origin DNN microservice for error correction that has multiple parameters separated by space. The first parameter “dnn_id: ” may represent the label for the identifier of the origin DNN microservice and has the parameter called “value”. The next parameter “destination_dnn_id” may represent the identifier of the destination DNN microservices and can have multiple identifier values. Finally, the parameter “algo” may represent the identifier whose value may be the name of the selected error correction algorithm an/or the associated parameters.

[0109] The novel message for step 3.11 can be sent as the following new SDP offer procedure’s capability attribute: a=time_horizon: dnn_id:value destination_dnn_id: id1 id2 id3 . . . idN changed_time_horizon: parameter 1 parameter2 . . . parameterN. The parameter “a=time_horizon: ” may be the new attribute representing the shorter time horizon selected by the DNN microservice that has multiple parameters separated by space. The first parameter “dnn_id: ” may represent the label for the identifier of the origin DNN microservice and has the parameter called “value”. The next parameter “destination_dnn_id” may represent the identifier of the destination DNN microservices and can have multiple identifier values. The parameter “changed_time_horizon” may represent the identifier for the proposed shorter time horizon and has multiple parameters (for example for the number of time slots and for their duration) as its value.

[0110] The novel message for step 3.13 can be sent as the following new SDP offer procedure’s capability attribute: a=selected_time_horizon: dnn_id:value destination_dnn_id: id1 id2 id3 . . . idN time_horizon: parameter1 parameter2 . . . parameterN. The parameter “a=selected_time_horizon: ” may be the new attribute representing the shorter time horizon selected by the origin DNN microservice that has multiple parameters separated by space. The first parameter “dnn_id: ” may represent the label for the identifier of the

origin DNN microservice and has the parameter called “value”. The next parameter “destination_dnn_id” may represent the identifier of the destination DNN microservices and can have multiple identifier values. Finally, the parameter “time_horizon” may represent the identifier for the selected shorter time horizon and has multiple parameters (for example for the number of time slots and for their duration) as its value.

[0111] Finally, the novel message for step 3.19 can be sent as the following new SDP offer procedure’s capability attribute: a=retrain_time: dnn_id:value destination_dnn_id: id1 id2 id3 . . . idN proposed_time: parameter1 parameter2 . . . parameterN. The parameter “a=retrain_time:” may be the new attribute representing the time at which the DNN needs to be retrained selected by the origin DNN microservice that has multiple parameters separated by space. The first parameter “dnn_id:” may represent the label for the identifier of the origin DNN microservice and has the parameter called “value”. The next parameter “destination_dnn_id” may represent the identifier of the destination DNN microservices and can have multiple identifier values. Finally, the parameter “proposed_time” may represent the identifier whose value may be the time at which retraining will be triggered. This time could be represented using multiple parameters—for example to specify a time interval rather than wall clock time.

[0112] FIG. 4 is a flowchart illustrating a representative method 400 implemented by a WTRU 102. Referring to FIG. 4, the representative method 400 may include, at block 410, sending, to a network entity, first information comprising first data, wherein the network entity comprises a plurality of artificial intelligence/machine learning (AIML) models.

[0113] At block 420, the representative method 400 may include receiving, from the network entity, second data wherein the second data are predicted data obtained from the aggregation of output data generated from the plurality of AIML models.

[0114] At block 430, the representative method 400 may include determining a prediction error based on the second data.

[0115] At block 440, the representative method 400 may include sending, to the network entity, based on a comparison of the prediction error and a threshold value, a list of error correction algorithms associated with the plurality of AIML models.

[0116] At block 450, the representative method 400 may include receiving, from the network entity, second information indicating a rank-ordered list of the error correction algorithms.

[0117] At block 460, the representative method 400 may include selecting an error correction algorithm based on the rank-ordered list of the error correction algorithms.

[0118] At block 470, the representative method 400 may include sending, to the network entity, third information indicating the selected error correction algorithm.

[0119] According to certain embodiments, the representative method 400 may comprise any of the following steps: (1) receiving, from the network entity, fourth information indicating a plurality of frequency values of prediction of data; (2) selecting a frequency value from the plurality of frequency values; and (3) sending, to the network entity, fifth information indicating the selected frequency value.

[0120] According to certain embodiments, the representative method 400 may comprise sending, to the network entity, information indicating a time instant to retrain the plurality of AIML models.

[0121] According to certain embodiments, one or more (e.g., each) AIML model of the plurality of AIML models is associated with a respective microservice running on the network entity.

[0122] According to certain embodiments, the network entity is included in an edge device.

[0123] According to certain embodiments, the first information comprises pre-processed data.

[0124] According to certain embodiments, the prediction data are associated with a prediction of a user action on the WTRU.

[0125] Although features and elements are provided above in particular combinations, one of ordinary skill in the art will appreciate that each feature or element can be used alone or in any combination with the other features and elements. The present disclosure is not to be limited in terms of the particular embodiments described in this application, which are intended as illustrations of various aspects. Many modifications and variations may be made without departing from its spirit and scope, as will be apparent to those skilled in the art. No element, act, or instruction used in the description of the present application should be construed as critical or essential to the invention unless explicitly provided as such. Functionally equivalent methods and apparatuses within the scope of the disclosure, in addition to those enumerated herein, will be apparent to those skilled in the art from the foregoing descriptions. Such modifications and variations are intended to fall within the scope of the appended claims. The present disclosure is to be limited only by the terms of the appended claims, along with the full scope of equivalents to which such claims are entitled. It is to be understood that this disclosure is not limited to particular methods or systems.

[0126] The foregoing embodiments are discussed, for simplicity, with regard to the terminology and structure of infrared capable devices, i.e., infrared emitters and receivers. However, the embodiments discussed are not limited to these systems but may be applied to other systems that use other forms of electromagnetic waves or non-electromagnetic waves such as acoustic waves.

[0127] It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to be limiting. As used herein, the term “video” or the term “imagery” may mean any of a snapshot, single image and/or multiple images displayed over a time basis. As another example, when referred to herein, the terms “user equipment” and its abbreviation “UE”, the term “remote” and/or the terms “head mounted display” or its abbreviation “HMD” may mean or include (i) a wireless transmit and/or receive unit (WTRU); (ii) any of a number of embodiments of a WTRU; (iii) a wireless-capable and/or wired-capable (e.g., tetherable) device configured with, inter alia, some or all structures and functionality of a WTRU; (iii) a wireless-capable and/or wired-capable device configured with less than all structures and functionality of a WTRU; or (iv) the like. Details of an example WTRU, which may be representative of any WTRU recited herein, are provided herein with respect to FIGS. 1A-ID. As another example, various disclosed embodiments herein supra and infra are described as

utilizing a head mounted display. Those skilled in the art will recognize that a device other than the head mounted display may be utilized and some or all of the disclosure and various disclosed embodiments can be modified accordingly without undue experimentation. Examples of such other device may include a drone or other device configured to stream information for providing the adapted reality experience.

[0128] In addition, the methods provided herein may be implemented in a computer program, software, or firmware incorporated in a computer-readable medium for execution by a computer or processor. Examples of computer-readable media include electronic signals (transmitted over wired or wireless connections) and computer-readable storage media. Examples of computer-readable storage media include, but are not limited to, a read only memory (ROM), a random access memory (RAM), a register, cache memory, semiconductor memory devices, magnetic media such as internal hard disks and removable disks, magneto-optical media, and optical media such as CD-ROM disks, and digital versatile disks (DVDs). A processor in association with software may be used to implement a radio frequency transceiver for use in a WTRU, UE, terminal, base station, RNC, or any host computer.

[0129] Variations of the method, apparatus and system provided above are possible without departing from the scope of the invention. In view of the wide variety of embodiments that can be applied, it should be understood that the illustrated embodiments are examples only, and should not be taken as limiting the scope of the following claims. For instance, the embodiments provided herein include handheld devices, which may include or be utilized with any appropriate voltage source, such as a battery and the like, providing any appropriate voltage.

[0130] Moreover, in the embodiments provided above, processing platforms, computing systems, controllers, and other devices that include processors are noted. These devices may include at least one Central Processing Unit (“CPU”) and memory. In accordance with the practices of persons skilled in the art of computer programming, reference to acts and symbolic representations of operations or instructions may be performed by the various CPUs and memories. Such acts and operations or instructions may be referred to as being “executed,” “computer executed” or “CPU executed.”

[0131] One of ordinary skill in the art will appreciate that the acts and symbolically represented operations or instructions include the manipulation of electrical signals by the CPU. An electrical system represents data bits that can cause a resulting transformation or reduction of the electrical signals and the maintenance of data bits at memory locations in a memory system to thereby reconfigure or otherwise alter the CPU’s operation, as well as other processing of signals. The memory locations where data bits are maintained are physical locations that have particular electrical, magnetic, optical, or organic properties corresponding to or representative of the data bits. It should be understood that the embodiments are not limited to the above-mentioned platforms or CPUs and that other platforms and CPUs may support the provided methods.

[0132] The data bits may also be maintained on a computer readable medium including magnetic disks, optical disks, and any other volatile (e.g., Random Access Memory (RAM)) or non-volatile (e.g., Read-Only Memory (ROM)) mass storage system readable by the CPU. The computer

readable medium may include cooperating or interconnected computer readable medium, which exist exclusively on the processing system or are distributed among multiple interconnected processing systems that may be local or remote to the processing system. It should be understood that the embodiments are not limited to the above-mentioned memories and that other platforms and memories may support the provided methods.

[0133] In an illustrative embodiment, any of the operations, processes, etc. described herein may be implemented as computer-readable instructions stored on a computer-readable medium. The computer-readable instructions may be executed by a processor of a mobile unit, a network element, and/or any other computing device.

[0134] There is little distinction left between hardware and software implementations of aspects of systems. The use of hardware or software is generally (but not always, in that in certain contexts the choice between hardware and software may become significant) a design choice representing cost versus efficiency tradeoffs. There may be various vehicles by which processes and/or systems and/or other technologies described herein may be effected (e.g., hardware, software, and/or firmware), and the preferred vehicle may vary with the context in which the processes and/or systems and/or other technologies are deployed. For example, if an implementer determines that speed and accuracy are paramount, the implementer may opt for a mainly hardware and/or firmware vehicle. If flexibility is paramount, the implementer may opt for a mainly software implementation. Alternatively, the implementer may opt for some combination of hardware, software, and/or firmware.

[0135] The foregoing detailed description has set forth various embodiments of the devices and/or processes via the use of block diagrams, flowcharts, and/or examples. Insofar as such block diagrams, flowcharts, and/or examples include one or more functions and/or operations, it will be understood by those within the art that each function and/or operation within such block diagrams, flowcharts, or examples may be implemented, individually and/or collectively, by a wide range of hardware, software, firmware, or virtually any combination thereof. In an embodiment, several portions of the subject matter described herein may be implemented via Application Specific Integrated Circuits (ASICs), Field Programmable Gate Arrays (FPGAs), digital signal processors (DSPs), and/or other integrated formats. However, those skilled in the art will recognize that some aspects of the embodiments disclosed herein, in whole or in part, may be equivalently implemented in integrated circuits, as one or more computer programs running on one or more computers (e.g., as one or more programs running on one or more computer systems), as one or more programs running on one or more processors (e.g., as one or more programs running on one or more microprocessors), as firmware, or as virtually any combination thereof, and that designing the circuitry and/or writing the code for the software and or firmware would be well within the skill of one of skill in the art in light of this disclosure. In addition, those skilled in the art will appreciate that the mechanisms of the subject matter described herein may be distributed as a program product in a variety of forms, and that an illustrative embodiment of the subject matter described herein applies regardless of the particular type of signal bearing medium used to actually carry out the distribution. Examples of a signal bearing medium include, but are not limited to, the following: a

recordable type medium such as a floppy disk, a hard disk drive, a CD, a DVD, a digital tape, a computer memory, etc., and a transmission type medium such as a digital and/or an analog communication medium (e.g., a fiber optic cable, a waveguide, a wired communications link, a wireless communication link, etc.).

[0136] Those skilled in the art will recognize that it is common within the art to describe devices and/or processes in the fashion set forth herein, and thereafter use engineering practices to integrate such described devices and/or processes into data processing systems. That is, at least a portion of the devices and/or processes described herein may be integrated into a data processing system via a reasonable amount of experimentation. Those having skill in the art will recognize that a typical data processing system may generally include one or more of a system unit housing, a video display device, a memory such as volatile and non-volatile memory, processors such as microprocessors and digital signal processors, computational entities such as operating systems, drivers, graphical user interfaces, and applications programs, one or more interaction devices, such as a touch pad or screen, and/or control systems including feedback loops and control motors (e.g., feedback for sensing position and/or velocity, control motors for moving and/or adjusting components and/or quantities). A typical data processing system may be implemented utilizing any suitable commercially available components, such as those typically found in data computing/communication and/or network computing/communication systems.

[0137] The herein described subject matter sometimes illustrates different components included within, or connected with, different other components. It is to be understood that such depicted architectures are merely examples, and that in fact many other architectures may be implemented which achieve the same functionality. In a conceptual sense, any arrangement of components to achieve the same functionality is effectively “associated” such that the desired functionality may be achieved. Hence, any two components herein combined to achieve a particular functionality may be seen as “associated with” each other such that the desired functionality is achieved, irrespective of architectures or intermedial components. Likewise, any two components so associated may also be viewed as being “operably connected”, or “operably coupled”, to each other to achieve the desired functionality, and any two components capable of being so associated may also be viewed as being “operably couplable” to each other to achieve the desired functionality. Specific examples of operably couplable include but are not limited to physically mateable and/or physically interacting components and/or wirelessly interactable and/or wirelessly interacting components and/or logically interacting and/or logically interactable components.

[0138] With respect to the use of substantially any plural and/or singular terms herein, those having skill in the art can translate from the plural to the singular and/or from the singular to the plural as is appropriate to the context and/or application. The various singular/plural permutations may be expressly set forth herein for sake of clarity.

[0139] It will be understood by those within the art that, in general, terms used herein, and especially in the appended claims (e.g., bodies of the appended claims) are generally intended as “open” terms (e.g., the term “including” should be interpreted as “including but not limited to,” the term

“having” should be interpreted as “having at least,” the term “includes” should be interpreted as “includes but is not limited to,” etc.). It will be further understood by those within the art that if a specific number of an introduced claim recitation is intended, such an intent will be explicitly recited in the claim, and in the absence of such recitation no such intent is present. For example, where only one item is intended, the term “single” or similar language may be used. As an aid to understanding, the following appended claims and/or the descriptions herein may include usage of the introductory phrases “at least one” and “one or more” to introduce claim recitations. However, the use of such phrases should not be construed to imply that the introduction of a claim recitation by the indefinite articles “a” or “an” limits any particular claim including such introduced claim recitation to embodiments including only one such recitation, even when the same claim includes the introductory phrases “one or more” or “at least one” and indefinite articles such as “a” or “an” (e.g., “a” and/or “an” should be interpreted to mean “at least one” or “one or more”). The same holds true for the use of definite articles used to introduce claim recitations. In addition, even if a specific number of an introduced claim recitation is explicitly recited, those skilled in the art will recognize that such recitation should be interpreted to mean at least the recited number (e.g., the bare recitation of “two recitations,” without other modifiers, means at least two recitations, or two or more recitations). Furthermore, in those instances where a convention analogous to “at least one of A, B, and C, etc.” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, and C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). In those instances where a convention analogous to “at least one of A, B, or C, etc.” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, or C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). It will be further understood by those within the art that virtually any disjunctive word and/or phrase presenting two or more alternative terms, whether in the description, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the terms, or both terms. For example, the phrase “A or B” will be understood to include the possibilities of “A” or “B” or “A and B.” Further, the terms “any of” followed by a listing of a plurality of items and/or a plurality of categories of items, as used herein, are intended to include “any of,” “any combination of,” “any multiple of,” and/or “any combination of multiples of” the items and/or the categories of items, individually or in conjunction with other items and/or other categories of items. Moreover, as used herein, the term “set” is intended to include any number of items, including zero. Additionally, as used herein, the term “number” is intended to include any number, including zero. And the term “multiple”, as used herein, is intended to be synonymous with “a plurality”.

[0140] In addition, where features or aspects of the disclosure are described in terms of Markush groups, those skilled in the art will recognize that the disclosure is also

thereby described in terms of any individual member or subgroup of members of the Markush group.

[0141] As will be understood by one skilled in the art, for any and all purposes, such as in terms of providing a written description, all ranges disclosed herein also encompass any and all possible subranges and combinations of subranges thereof. Any listed range can be easily recognized as sufficiently describing and enabling the same range being broken down into at least equal halves, thirds, quarters, fifths, tenths, etc. As a non-limiting example, each range discussed herein may be readily broken down into a lower third, middle third and upper third, etc. As will also be understood by one skilled in the art all language such as “up to,” “at least,” “greater than,” “less than,” and the like includes the number recited and refers to ranges which can be subsequently broken down into subranges as discussed above. Finally, as will be understood by one skilled in the art, a range includes each individual member. Thus, for example, a group having 1-3 cells refers to groups having 1, 2, or 3 cells. Similarly, a group having 1-5 cells refers to groups having 1, 2, 3, 4, or 5 cells, and so forth.

[0142] Moreover, the claims should not be read as limited to the provided order or elements unless stated to that effect. In addition, use of the terms “means for” in any claim is intended to invoke 35 U.S.C. § 112, ¶ 6 or means-plus-function claim format, and any claim without the terms “means for” is not so intended.

REFERENCES

[0143] The contents of each of the following references are incorporated by reference herein:

- [0144] [1] PCT Publication WO 2023014901 A1.
- [0145] [2] Coulouris, et al., “Distributed Systems Concepts and Designs”, 5th Edition, 2012.
- [0146] [3] Culler, et al., “Parallel computer architecture: a hardware/software approach”, Gulf Professional Publishing, 1999.
- [0147] [4] Foster, Ian, “Designing and building parallel programs”, 1995.
- [0148] [5] LEE, et al., “Outatime Using Speculation to Enable Low-Latency Continuous Interaction for Mobile Cloud Gaming”, MobiSys '15: Proceedings of the 13th Annual International Conference on Mobile Systems, Applications and Services, May 2015.
- [0149] [6] <https://datatracker.ietf.org/doc/html/rfc5939>
- [0150] [7] <https://datatracker.ietf.org/doc/html/rfc6871>
- [0151] [8] <https://datatracker.ietf.org/doc/html/rfc3311>
- [0152] [9] <https://datatracker.ietf.org/doc/html/rfc4566>

What is claimed is:

1. A method implemented by a wireless transmit/receive unit (WTRU), the method comprising:

- sending, to a network entity, first information comprising first data, wherein the network entity comprises a plurality of artificial intelligence/machine learning (AIML) models;

- receiving, from the network entity, second data, wherein the second data are predicted data obtained from the aggregation of output data generated from the plurality of AIML models;

- determining a prediction error based on the second data;
- sending, to the network entity, based on a comparison of the prediction error and a threshold value, a list of error correction algorithms associated with the plurality of AIML models;

- receiving, from the network entity, second information indicating a rank-ordered list of the error correction algorithms;

- selecting an error correction algorithm based on the rank-ordered list of the error correction algorithms; and
- sending, to the network entity, third information indicating the selected error correction algorithm.

2. The method of claim 1, further comprising:

- receiving, from the network entity, fourth information indicating a plurality of frequency values of prediction of data;

- selecting a frequency value from the plurality of frequency values; and

- sending, to the network entity, fifth information indicating the selected frequency value.

3. The method of claim 1, further comprising: sending, to the network entity, information indicating a time instant to retrain the plurality of AIML models.

4. The method of claim 1, wherein each AIML model of the plurality of AIML models is associated with a respective microservice running on the network entity.

5. The method of claim 1, wherein the network entity is included in an edge device.

6. The method of claim 1, wherein the first information comprises pre-processed data.

7. The method of claim 1, wherein the prediction data are associated with a prediction of a user action on the WTRU.

8. A wireless transmit/receive unit (WTRU) comprising circuitry, including a transmitter, a receiver, a processor and memory, the WTRU configured to:

- send, to a network entity, first information comprising first data, wherein the network entity comprises a plurality of artificial intelligence/machine learning (AIML) models;

- receive, from the network entity, second data, wherein the second data are predicted data obtained from the aggregation of output data generated from the plurality of AIML models;

- determine a prediction error based on the second data;

- send, to the network entity, based on a comparison of the prediction error and a threshold value, a list of error correction algorithms associated with the plurality of AIML models;

- receive, from the network entity, second information indicating a rank-ordered list of the error correction algorithms;

- select an error correction algorithm based on the rank-ordered list of the error correction algorithms; and

- send, to the network entity, third information indicating the selected error correction algorithm.

9. The WTRU of claim 8, further configured to:

- receive, from the network entity, fourth information indicating a plurality of frequency values of prediction of data;

- select a frequency value from the plurality of frequency values; and

- send, to the network entity, fifth information indicating the selected frequency value.

10. The WTRU of claim 8, further configured to: send, to the network entity, information indicating a time instant to retrain the plurality of AIML models.

11. The WTRU of claim 8, wherein each AIML model of the plurality of AIML models is associated with a respective microservice running on the network entity.

12. The WTRU of claim 8, wherein the network entity is included in an edge device.

13. The WTRU of claim 8, wherein the first information comprises pre-processed data.

14. The WTRU of claim 8, wherein the prediction data are associated with a prediction of a user action on the WTRU.

15. A non-transitory storage medium having instructions which, when executed, cause a device to perform the method of claim 1.

16. A non-transitory storage medium having instructions which, when executed, cause a device to perform the method of claim 2.

17. A non-transitory storage medium having instructions which, when executed, cause a device to perform the method of claim 3.

18. A non-transitory storage medium having instructions which, when executed, cause a device to perform the method of claim 4.

19. A non-transitory storage medium having instructions which, when executed, cause a device to perform the method of claim 5.

20. A non-transitory storage medium having instructions which, when executed, cause a device to perform the method of claim 1, wherein at least one of:

the first information comprises pre-processed data; and
the prediction data are associated with a prediction of a user action on the WTRU.

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